

Matrix Rings and the McCoy Property: A Proof of Conjecture 00000002226

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Abstract

For a nonzero commutative ring R and $n \geq 1$, the matrix ring $M_n(R)$ has the McCoy property if and only if $n = 1$. The case $n = 1$ is McCoy's theorem. For $n \geq 2$ an explicit pair of linear polynomials shows that $M_n(R)$ is neither right nor left McCoy. The equivalence and both one-sided versions are verified in Lean.

Definitions and reading of the source. Rings are associative with 1, and $A[x]$ is the polynomial ring in a central indeterminate. Following Nielsen [2], a ring A is *right McCoy* if for all nonzero $f, g \in A[x]$ with $fg = 0$ there is a nonzero $r \in A$ with $fr = 0$. It is *left McCoy* if under the same hypothesis there is a nonzero $s \in A$ with $sg = 0$. It is *McCoy* if it is both. This is the property in the source: the coefficients of a zero-divisor polynomial are annihilated by one nonzero element.

The source considers $M_n(R)$ over a commutative ring R ; the Chinese version states the same equivalence. We take R nonzero and $n \geq 1$. These are the standing conventions for a matrix ring; without them $M_n(R)$ is the zero ring, which is McCoy vacuously because it has no nonzero polynomials. The last section returns to this point.

Theorem 1. *Let R be a nonzero commutative ring and $n \geq 1$. Then $M_n(R)$ is McCoy if and only if $n = 1$. The same equivalence holds for the right McCoy property alone and for the left McCoy property alone.*

The theorem follows from the next two lemmas, since $M_1(R) \cong R$ and both properties are invariant under ring isomorphism.

Lemma 1 (McCoy [1]). *Every commutative ring is McCoy.*

Proof. Let $f, g \in R[x]$ be nonzero with $fg = 0$. Among all nonzero h with $fh = 0$ choose one of least degree m , and write $f = \sum_i a_i x^i$ and $h = \sum_{j \leq m} b_j x^j$ with $b_m \neq 0$. If $a_i h = 0$ for every i , then $a_i b_m = 0$ for every i , so $b_m f = 0$ with $b_m \neq 0$. Otherwise let l be the largest index with $a_l h \neq 0$. Then $0 = fh = (a_0 + \cdots + a_l x^l)h$, and the coefficient of x^{l+m} gives $a_l b_m = 0$. Hence $a_l h$ is nonzero of degree less than m , and $f \cdot a_l h = a_l fh = 0$. This contradicts the choice of h . So some nonzero $r \in R$ satisfies $fr = rf = 0$. Applying this to $gf = 0$ gives a nonzero s with $sg = 0$. $\square$

Lemma 2. *Let S be a nonzero ring, not necessarily commutative, and $n \geq 2$. Then $M_n(S)$ is neither right McCoy nor left McCoy.*

Proof. Let E_{ab} be the matrix units, so $E_{ab}E_{cd} = \delta_{bc}E_{ad}$, and let I be the identity matrix. Put

$$f = (I - E_{22}) + E_{12}x, \quad g = E_{21} - E_{11}x.$$

Both are nonzero because $E_{12} \neq 0$ and $E_{21} \neq 0$. The three coefficients of fg are

$$\begin{aligned} (I - E_{22})E_{21} &= E_{21} - E_{21} = 0, \\ E_{12}E_{21} - (I - E_{22})E_{11} &= E_{11} - E_{11} + 0 = 0, \\ -E_{12}E_{11} &= 0, \end{aligned}$$

so $fg = 0$. Let $r \in M_n(S)$ with $fr = 0$. Then $(I - E_{22})r = 0$ and $E_{12}r = 0$. The first equation gives $r = E_{22}r$, and therefore

$$r = E_{22}r = E_{21}E_{12}r = 0.$$

Thus no nonzero constant annihilates f on the right, and $M_n(S)$ is not right McCoy.

For the left property put

$$f' = E_{12} - E_{11}x, \quad g' = (I - E_{22}) + E_{21}x.$$

Both are nonzero. The coefficients of $f'g'$ are $E_{12}(I - E_{22}) = 0$, $E_{12}E_{21} - E_{11}(I - E_{22}) = E_{11} - E_{11} = 0$ and $-E_{11}E_{21} = 0$, so $f'g' = 0$. If $sg' = 0$, then $s(I - E_{22}) = 0$ and $sE_{21} = 0$, hence $s = sE_{22} = sE_{21}E_{12} = 0$. Thus $M_n(S)$ is not left McCoy. $\square$

Formal verification and scope

The Lean definitions `RightMcCoy`, `LeftMcCoy` and `McCoy` are stated for an arbitrary ring A and Mathlib's polynomial ring $A[X]$, whose indeterminate is central. They are the definitions above, with `C r` the constant polynomial.

The first lemma is proved in `rightMcCoy_of_comm` and `leftMcCoy_of_comm` from Mathlib's McCoy theorem

`Polynomial.nmem_nonZeroDivisors_iff.`

The explicit ring isomorphism `oneByOne` between $M_1(R)$ and R , with the two transfer lemmas named `of_ringEquiv`, gives `mcCoy_one`.

`not_rightMcCoy_matrix` and `not_leftMcCoy_matrix` prove the second lemma for matrices over any nonzero ring indexed by any finite type with two distinct indices. They use the polynomials displayed above, built from Mathlib's matrix units; the products, the nonvanishing of f and g , and the vanishing of every annihilating constant are all proved. The main theorem is `matrix_mcCoy_iff`: for a nontrivial commutative ring and $1 \leq n$,

$$\text{McCoy (Matrix (Fin n) (Fin n) R)} \leftrightarrow n = 1.$$

`matrix_rightMcCoy_iff` and `matrix_leftMcCoy_iff` are the one-sided versions. No `sorry`, `native_decide` or custom axiom occurs.

The hypotheses $R \neq 0$ and $n \geq 1$ are the only restriction. `mcCoy_of_subsingleton` records why they are needed: a zero ring is McCoy vacuously, so for $R = 0$ or $n = 0$ the ring $M_n(R)$ is McCoy for a trivial reason. Under the usual convention that the coefficient ring of a matrix ring is nonzero, the source statement is proved in full. A reader who admits the zero ring should read the result as: the equivalence holds exactly for nonzero R .

Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`. Its repository path is

`conjectures/00000002226.md.`

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b.`

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean.`

The included public-Git manifest locks all dependency revisions. The `verification/` directory records the fresh-build logs, theorem-axiom reports, artifact hashes, review and final PDF inspection. No supplementary numerical computation is required.

References

- [1] N. H. McCoy, Remarks on divisors of zero, *Amer. Math. Monthly* 49 (1942), 286–295.
- [2] P. P. Nielsen, Semi-commutativity and the McCoy condition, *J. Algebra* 298 (2006), 134–141.